

TARUN REDDY NERELLA

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AI DATA ENGINEER | Databricks, Spark, Delta Lake, Cloud Data Platforms, RAG, Data Observability

PROFESSIONAL SUMMARY

AI Data Engineer building healthcare data pipelines on Databricks, Spark, Azure Data Factory, Azure Synapse, Azure Data Lake Storage and Delta Lake. Contributed to Synapse to Databricks migration across bronze, silver and gold layers for production processes reaching 2B+ rows. Extends data engineering into RAG, vector databases, prompt engineering, local LLMs, data observability and agent-assisted workflows.

CERTIFICATIONS

Microsoft Certified: [Azure Data Engineer Associate](#) | Apr 2024

[Databricks Certified Data Engineer Associate](#) | May 2024

[Databricks Certified Data Engineer Professional](#) | Jun 2024

[Databricks Certified Generative AI Engineer Associate](#) | May 2026

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Java, JavaScript, C++

Data Platforms: Databricks, Spark, Delta Lake, Unity Catalog, Lakeflow, Snowflake, Snowpipe, Azure Data Factory, Synapse, ADLS

Cloud and Streaming: Azure, GCP BigQuery, Dataflow, Pub/Sub, Cloud Run, GCS, Kafka, AWS

Data Engineering and Observability: ETL/ELT, medallion architecture, batch/streaming, data contracts, data quality, schema evolution, lineage, SLA monitoring

AI Engineering: RAG, embeddings, vector databases, semantic search, chunking, prompt engineering, LLM evaluation, local LLMs

Agent Tools: Agent Bricks, ReAct agents, multi-agent workflows, REST APIs, CLI, MCP, Claude Code, Codex, Cursor

DevOps/ML: Azure DevOps, Docker, CI/CD, Git, Jenkins, TensorFlow, PyTorch, OpenCV, YOLO, Hugging Face, Ollama, Plotly/Dash

PROFESSIONAL EXPERIENCE

Data Engineer | Hexaware Technologies | Jul 2024 to Present

- Contributed to Synapse to Databricks migration for healthcare workflows across bronze, silver and gold layers, translating orchestration, validation and table logic into lakehouse patterns.
- Build and support Databricks and Azure Data Factory ingestion pipelines in Scala/Python for healthcare data reaching 2B+ rows, with parameterized jobs, dependencies and production checks.
- Develop validation, reconciliation and triage workflows for CMS, HEDIS, BITS, PerformRx, fee schedules, premium files and healthcare EDI 834/820 feeds.
- Built file observability for hundreds of daily inbound files, each with millions of records, tracking source/domain readiness, load status, metadata gaps and recovery actions.
- Improved Spark runtime 30 to 40 percent by tuning partitioning, workflow timing, job dependencies and execution plans.
- Added validation, reconciliation and reprocessing controls to protect critical tables, reduce incomplete loads and cut manual triage.
- Use Databricks Agent Bricks and coding agents to debug notebooks, inspect code paths and investigate production issues.
- Coordinate release validation, monitoring, incident response, stakeholder updates and Azure DevOps production runbooks.

Graduate Student Assistant | CDC, University of Colorado Denver | Denver, CO | Jan 2023 to May 2024

- Supported ML education research through Python experiments, reproducible notebooks, model checks and result validation.
- Mentored 20+ students on Python, debugging, data structures and code quality through technical guidance and code reviews.

Data Engineer | 9th Networks Inc. | Aug 2020 to Aug 2022

- Built batch and streaming pipelines for utility telemetry analytics, including smart meter interval reads, device events, outages, restorations and asset operations data.
- Loaded AMI and MDM extracts into Snowflake with Snowpipe, schema checks, audit fields, duplicate handling, reject tracking, load history and reconciliation evidence.
- Developed Kafka pub/sub flows for meter status, outages, restorations, voltage anomalies and operational alerts used by downstream dashboards and exception views.
- Modeled curated interval usage, meter events, outages, service points and asset hierarchy tables for utility reporting.
- Created Python reconciliation checks for counts, nulls, invalid fields, timestamps, late files, duplicate events and volume drift.
- Built Plotly/Dash dashboards for file arrival, load status, failures, processing latency and anomaly review.
- Standardized Docker/CI/CD deployment patterns for data services and dashboards to improve release repeatability.

Full-Stack Developer | IIIT Hyderabad | Hyderabad, India | Jun 2020 to Aug 2020

- Built React/Node.js web app with modular front-end flows, backend APIs and D3.js visualizations for interactive user workflows.
- Tuned API and server behavior for responsiveness, load performance and modular feature expansion.

PROJECTS

Data Contract Observability System | Azure Databricks, Delta Lake, Unity Catalog, Lakeflow | 2026

- Built a data contract observability workflow for schema drift, type changes, null spikes, freshness misses, row count drift and SLA breaks across downstream Delta tables.
- Layered contracts, owners, severity scoring, lineage impact, change history and remediation status over Unity Catalog metadata.
- Runs after ingestion jobs, ADLS events and release checks, publishing owner-ready alerts with impacted tables, sample records, suggested fixes and follow-up context.

HomeOps AI Native Data Pipeline | GCP, Cloud Run, PostgreSQL, pgvector, LangGraph, MCP | 2025

- Built metadata-driven IoT ingestion pipeline that detects new devices, infers schemas, validates events and creates reusable source adapters without manual onboarding.
- Uses Cloud Run, PostgreSQL, pgvector, LangGraph and REST/MCP tools for source discovery, metadata approval, audit history and natural language operations.
- Uses vector search over device docs and prior adapters to recommend schemas, transformations and validation rules.

ATRIA Human Agent CRM Workflow Platform | Next.js, NestJS, PostgreSQL, Redis, Ollama, BYOK | 2026

- Built human-agent CRM workflows for agent-assisted tasks, approvals, trust scoring, relationship follow-ups and cost controls.
- Routes OpenAI, Anthropic and Ollama through PostgreSQL row-level security, Redis queues, BYOK keys and audit trails.
- Captures agent actions, human overrides and approval decisions for explainable, reviewable and cost-aware automation.

Codex Council Agent Collaboration Plugin | Claude Code, Codex CLI, Node.js, JSONL | 2026

- Built Claude Code plugin coordinating Claude, subagents and Codex through structured task packets, critique loops and transcripts.
- Implemented slash commands, lock-safe JSONL logs, schema validation, secret scanning and automated tests across repeatable agent collaboration paths.

Local Meeting Intelligence Agent | Ollama, Obsidian, Google Calendar | 2025

- Built local LLM meeting agent that converts transcripts into Obsidian graph notes for participants, decisions, actions and owners.
- Combines local Ollama summaries with Google Calendar context to identify follow-ups and prepare time-zone-aware meeting recommendations with context.

ReAct Fitness Recommendation Agent | APIs, Email Automation, Reasoning Loops | 2026

- Built ReAct agent that analyzes 30 days of fitness exports, workout history, recent training load and recovery constraints before recommending the next workout.
- Emails the recommendation with reasoning and stores decisions, constraints and history so the workflow remains auditable.

Applied ML and Software Portfolio | TensorFlow, OpenCV, YOLO, Kafka, Spark, Django | 2021 to 2023

- Built applied ML systems for waste segregation, road sign detection, assistive vision, news classification and maze solving using TensorFlow, OpenCV, YOLOv5/v7 and NLP/CV workflows.
- Built Django polyclinic, Flutter/Firebase e-commerce and React/Node/D3 apps with auth, dashboards, APIs and product workflows.
- Used Kafka, Spark and Python pipelines for ingestion, model training, inference, dashboards and user-facing applications.

EDUCATION

Master of Science, Computer Science | University of Colorado Denver | Denver, CO | GPA: 3.8 | Aug 2022 to May 2024

Relevant Coursework: Big Data Systems, Artificial Intelligence, Deep Learning, Computer Vision, Machine Learning

Bachelor of Technology, Computer Science and Engineering

JB Institute of Engineering and Technology | Hyderabad, India | Aug 2018 to Jun 2022

Relevant Coursework: Data Structures, Operating Systems, Computer Networks, Database Management Systems

RESEARCH AND TRAINING

Publication: [Hazardous Asteroid Prediction using XGBoost](#) | IRJET, Sep 2021

Research Focus: XGBoost classifier feature selection for hazardous asteroid prediction

Summer Analytics 2021: IIT Guwahati | ML algorithms, statistical modeling and applied analytics

ML Program: E-Cell IIT Kharagpur | Predictive modeling, business analytics and applied machine learning